

USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech. (CST) (CBCS) Degree Examinations

January 2023

22CST123 – COMPUTATIONAL STRUCTURAL MECHANICS

Duration: 3 Hour

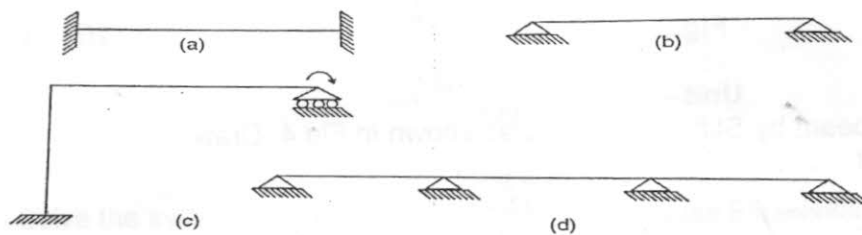
Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Assume suitable data if anything missing.

Unit – I

1. a) State the difference between flexibility method and stiffness method.
- b) Prove that relationship between flexibility and stiffness are inverse of each other.
- c) Explain degrees of freedom. Find the kinematic indeterminacy of following structures:



- d) Explain a coordinate system in structures with example.

2. a) Develop stiffness matrix for a given structure with coordinates as shown in Fig 2(a).

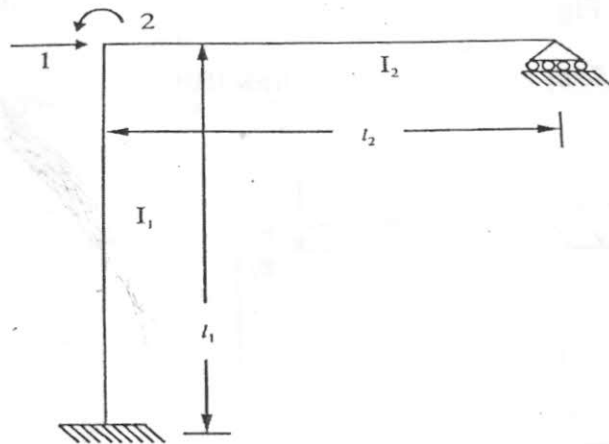


Fig 2(a).

- b) Develop a flexibility matrix for a given structure with coordinates as shown in Fig 2(b).



Fig 2(b).

3. Analyze the truss by flexibility method as shown in Fig.3.

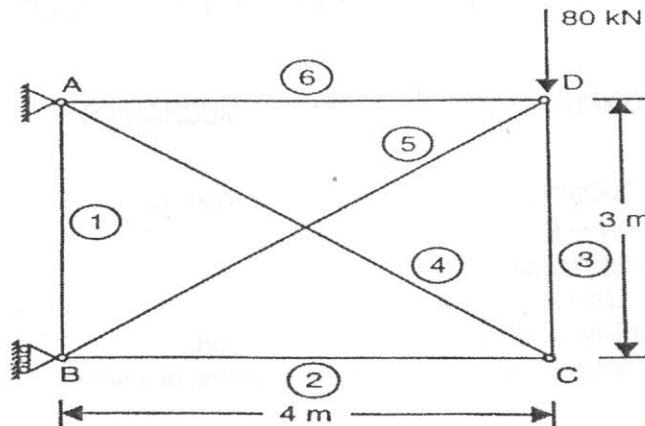


Fig.3.

20 L4 2 1,2

Unit – II

4. Analyze the continuous beam by Stiffness method as shown in Fig.4. Draw BMD. Take EI is constant.

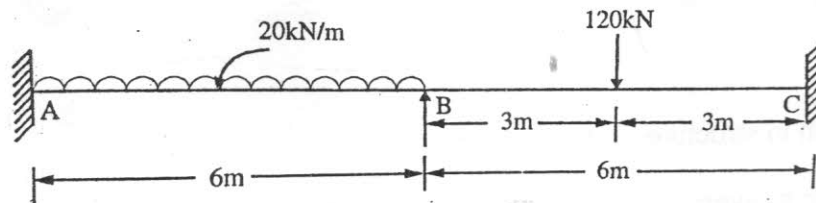


Fig.4

20 L4* 3 1,2

5. Analyze the frame as shown in Fig.5. by stiffness method. Draw BMD.

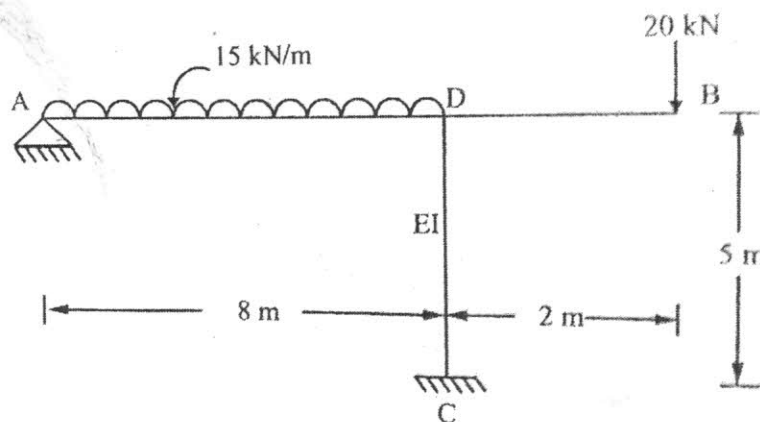


Fig.5

20 L4 3 1,2

6. Analyze the truss as shown in Fig.6 by direct stiffness method. Draw BMD.
Take $A=1000\text{mm}^2$ and $E=300\text{kN/mm}^2$.

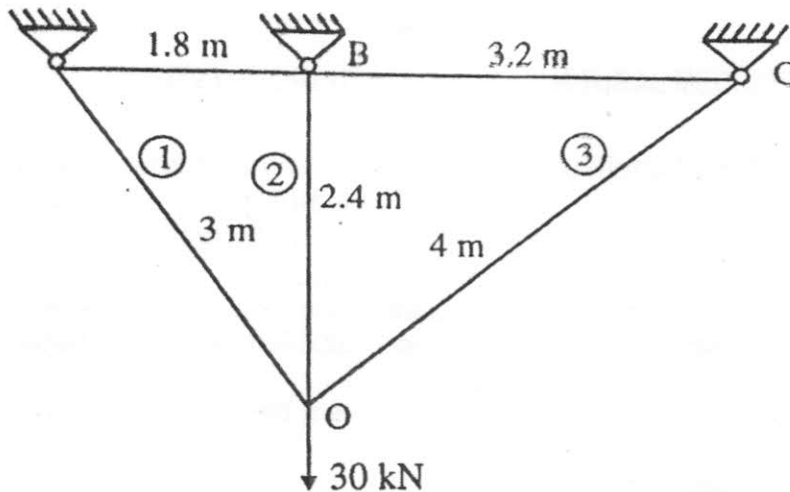


Fig.6

20 L4 4 1,2

Unit – III

7. Solve the system of equations mentioned using gauss Elimination method.
 $x+y+z = 2$
 $x+2y+3z = 5$
 $2x+3y+4z = 11$
8. The top and bottom surfaces of continuous beam shown in Fig. 8 are heated to 20°C and 10°C respectively. Solve the problem using stiffness approach.
 Take $E = 2 \times 10^5 \text{N/mm}^2$,
 Coefficient of temperature $= 1.2 \times 10^{-5}/^\circ\text{C}$.
 $I_{AB} = 2 \times 10^8 \text{mm}^4$, $I_{BC} = 5 \times 10^7 \text{mm}^4$ and $h = 400\text{mm}$.

20 L3 5 1,2

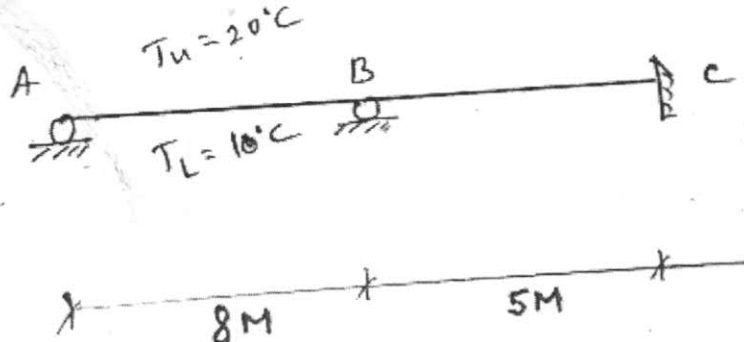


Fig. 8

20 L4 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
